

Juhen Fashikha Wildan

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EDUCATION

Universitas Airlangga – Surabaya, Indonesia

Aug 2022 - Jun 2026

Bachelor of Engineering (S.T.) in Robotics and Artificial Intelligence Engineering | GPA 3.72/4.00

Relevant coursework: Computer Vision, Deep Learning, Robotics, Automated System

PROFESSIONAL EXPERIENCE

PT. Yamaha Musical Products Indonesia – Pasuruan, Indonesia

Feb 2025 - Aug 2025

Process Automation Engineer (Intern)

- Follow-Me Tour Robot: Acted as primary ROS 2 programmer and electrical assistant for a factory-tour sound robot; integrated LiDAR, UWB, and Ultrasonic sensors for real-time human-following and autonomous tracking.
- Material Washing Monitoring: Programmed a Python GUI integrating a YOLO model to automatically track material dipping via CCTV streams; automated duration-tracking logic and CSV data logging to streamline company operational reporting.
- Holabot Pudu Robot Monitoring: Developed a real-time monitoring system deploying YOLO object detection across 3 distinct CCTV streams to identify delivery robot arrivals; integrated database tracking to trigger station sirens and live screen alerts, accelerating operator response times.

PT. Adhikara Wiyasa Gani – Surabaya, Indonesia

Sept 2024 - Jan 2025

Computer Vision Engineer (Intern)

- Visual Inspection System: Designed an automated QR detection system and Tkinter GUI using Python and OpenCV; integrated a Huateng Vision industrial camera via MindVision SDK (MVSDK) for real-time exposure/ROI control and Otsu's thresholding.
- Empirical Testing & Optimization: Executed a systematic experimental matrix across 24 distinct configurations (5–20 cm heights, 0.5–3.0 cm QR sizes) with 30 samples per test case, achieving 95% detection accuracy under optimal setups.

ORGANIZATION

Kombo Unair – Surabaya, Indonesia

Apr 2024 - Feb 2025

Research & Development Team

- Omnidirectional Kinematics: Researched 3-wheel omnidirectional drive mechanisms and mobile robot kinematics to analyze platform maneuverability and structural design constraints.
- Autonomous Navigation: Investigated autonomous navigation stacks, focusing on the A* algorithm for global path planning and the DWB planner for real-time local obstacle avoidance and trajectory control.

PUBLICATION

- Co-Author (2nd Author): "Multi-microcontroller system for Mecanum robots with gripper-shooter mechanisms" – IAES International Journal of Robotics and Automation (IJRA), 2025. Indexed in Scopus Q4 and Sinta 2.
- Lead Author (1st Author): "A Review: Artificial Intelligence Related to Agricultural Equipment Integrated with the Internet of Things" – Journal of Advanced Technology and Multidiscipline (JATM). Garnered 950 full-article views.

SKILLS

Programming languages: Python, C++, MATLAB, HTML.

Software & Frameworks: OpenCV, TensorFlow, PyTorch, YOLO, Machine Learning, Deep Learning, GitHub, Docker, ROS (Robot Operating System).

Core Competencies: Computer Vision, Camera Calibration, Image Pre-processing, Object Detection/Tracking, QR/Barcode Scanning, Real-Time Monitoring, Automated Systems, Experimental Analysis.

Languages: Indonesia (Native), English (Intermediate).